If the world were, however, made of things, what would these things be? The atoms, which we have discovered to be made up in turn of smaller particles? The elementary particles, which, as we have discovered, are nothing other than the ephemeral agitations of a field? The quantum fields, which we have found to be little more than codes of a language with which to speak of interactions and events? We cannot think of the *physical* world as if it were made of things, of entities. It simply doesn't work.

What works instead is thinking about the world as a network of events. Simple events, and more complex events that can be disassembled into combinations of simpler ones. A few examples: a war is not a thing, it's a sequence of events. A storm is not a thing, it's a collection of occurrences. A cloud above a mountain is not a thing, it is the condensation of humidity in the air that the wind blows over the mountain. A wave is not a thing, it is a movement of water, and the water that forms it is always different. A family is not a thing, it is a collection of relations, occurrences, feelings. And a human being? Of course it's not a thing; like the cloud above the mountain, it's a complex process, where food, information, light, words, and so on enter and exit. . . . A knot of knots in a network of social relations, in a network of chemical processes, in a network of emotions exchanged with its own kind.

For a long time, we have tried to understand the world in terms of some primary *substance*. Perhaps physics, more than any other discipline, has pursued this primary substance. But the more we have studied it, the less the world seems comprehensible in terms of something that *is*. It seems to be a lot more intelligible in terms of relations between events.

The words of Anaximander quoted in the first chapter of this book invited us to think of the world "according to the order of time." If we do not assume a priori that we know *what* the order of time is—if we do not, that is, presuppose that it is the linear and universal order that we are accustomed to—Anaximander's exhortation remains valid: we understand the world by studying change, not by studying things.

Those who have neglected this good advice have paid a heavy price for it. Two of the greats who fell into this error were Plato and Kepler, both curiously seduced by the same mathematics.

In the *Timaeus*, Plato has the excellent idea of attempting to translate into mathematics the physics insights gained by atomists such as Democritus. But he goes about it the wrong way: he tries to write the mathematics of the *shape* of atoms, rather than the mathematics of their *movements*. He allows himself to be fascinated by a mathematical theorem which establishes that there are five—and *only* five—regular polyhedra, these ones: